

EchoCode

Business Revenue Plan | Cohort and Campus

Voice-first AI pair programming for people who read code they did not write

\$0.12 cost per question (post-optimisation)	83% gross margin cohort deal	65% gross margin campus deal	\$469k gross profit year 3
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Executive summary

EchoCode sells to institutions, not to students. Students get a permanent free tier and act as the distribution channel; the invoice goes to the organisation whose measured outcome improves. Two customer types carry the plan. **Bootcamps** buy per cohort, decide in weeks, and are scored on completion rate. **University departments** buy per seat per year, decide in months, and represent the larger contract.

The model is built on a verified cost base rather than an assumption. AssemblyAI's Voice Agent API is billed per second of session duration at \$4.50/hour, and code-card generation through the LLM Gateway costs \$0.0014 per card - approximately 0.2% of the cost of goods sold. Cost of goods is therefore almost entirely voice session time, it is metered per install, and it is capped in product code.

The central insight. The cost driver is idle session time, not question volume. A question consumes 45 seconds of speech but bills up to three minutes of open session. Reducing the idle timeout from three minutes to sixty seconds takes cost per question from \$0.225 to \$0.131 - a single constant worth more to gross margin than any change to price.

Target customers

	Cohort (bootcamps)	Campus (departments)
Buyer	Programme director	Department head or course lead
Unit sold	Per learner, per cohort	Per seat, per academic year
Price	\$25 per learner	\$36 per seat
Typical deal	30 learners = \$750	500 seats = \$18,000
Sales cycle	2-6 weeks	3-9 months
Their KPI	Completion rate	Pass rate, teaching-hours load
Role in plan	Beachhead: revenue now, proof fast	Scale: larger contracts, renewals

Bootcamps come first. They decide quickly, their economics are transparent, and dropout is an event they can already price. University departments are the larger prize but cannot be closed inside a first commercial year, so they are sold into course-by-course rather than campus-wide to bypass procurement.

Pricing architecture

Every allowance is denominated in **session minutes**, the unit AssemblyAI actually bills, rather than spoken-audio minutes. Allowances are pooled across a cohort or department so that pricing reflects the cohort average instead of the heaviest individual user.

Tier	Price	Included	Max COGS	Margin
Free	\$0	10 questions / month	\$1.20 / mo	Acquisition
Cohort	\$25 / learner	~36 questions / 12 weeks	\$4.32	83%
Campus	\$36 / seat / yr	120,000 session-min pooled	\$12.60	65%
Pro	\$20 / month	60 questions / month	\$7.20	64%
Overage	\$0.15 / session-min	Beyond pooled allowance	\$0.075	50%

Unit economics

Cost component	Rate	Basis
Voice Agent session	\$0.075 / min	Per second of open session (\$4.50/hr)
Code card (LLM Gateway)	\$0.0014 / card	qwen3.5-4b-32k-fast, ~12k tokens in
Infrastructure	~\$360 / year	Vercel Pro and Upstash Redis, all customers

Cost per question, by idle timeout

A question is 45 seconds of speech. The rest of the billed time is the session sitting open while the developer reads and thinks, plus the idle tail before it closes.

Idle timeout	Billed min / question	Cost / question	Status
3 minutes	3.00	\$0.225	As shipped today
60 seconds	1.75	\$0.131	Planning assumption
30 seconds	1.25	\$0.094	Free tier target
Audio only (floor)	0.75	\$0.056	Theoretical minimum

Responsiveness becomes a feature we sell. The idle timeout is tiered by plan - 30 seconds on Free, 60-90 seconds on Cohort and Campus, 3 minutes on Pro. Paying customers get instant follow-up questions; the free tier costs us the least it can. The same constant that controls gross margin also differentiates the paid product.

Deal-level profit and loss

Cohort deal - one bootcamp intake of 30 learners

Line	Calculation	Amount
Revenue	30 learners x \$25	\$750.00
Voice COGS	1,080 questions x \$0.12	(\$129.60)
Code cards	~324 cards x \$0.0014	(\$0.45)
Gross profit	83% margin	\$619.95

A bootcamp partner running four intakes a year is worth **\$3,000 in revenue** and **\$2,480 in gross profit**, won with roughly five hours of founder-led selling.

Campus deal - 500 seats at \$36

Line	Calculation	Amount
Revenue	500 seats x \$36	\$18,000
Voice COGS	52,500 questions x \$0.12	(\$6,300)
Code cards	~15,750 cards x \$0.0014	(\$22)
Gross profit	65% margin	\$11,678

Adoption sensitivity - the risk is success

Blended questions per seat equal the adoption rate multiplied by the usage of those who adopt. Because voice is a real cost of goods, high adoption compresses margin. This is the question an experienced judge or investor asks first, and the pooled allowance with paid overage is the answer to it.

Adoption	Blended q / seat / yr	Voice COGS	Gross profit	Margin
40% - light	70	\$4,200	\$13,800	77%
60% - expected	105	\$6,300	\$11,700	65%
80% - mandated	160	\$9,600	\$8,400	47%
100% - heavy	240	\$14,400	\$3,600	20%

Downside is contractually capped. The 500-seat licence includes 120,000 pooled session minutes. Beyond that, usage bills at \$0.15 per session minute - double our cost. The heavy-adoption row therefore does not erode margin in practice; it converts into expansion revenue, and the deals that grow fastest are the ones that grow without a new sale.

Prices and deal counts in this document are modelled hypotheses, not signed contracts. The cost base - \$0.075 per session minute and \$0.0014 per code card - is verified against live AssemblyAI pricing.

Three-year projection

Bottom-up from deal counts, at the \$0.12 planning cost per question. Free-tier consumption is carried as a marketing cost at 15% monthly-active of cumulative sign-ups.

	Year 1	Year 2	Year 3
Bootcamp partners	10	30	60
Cohorts delivered	40	120	240
Campus contracts	2	8	25
Pro subscribers	50	200	600
Cohort revenue	\$30,000	\$90,000	\$180,000
Campus revenue	\$36,000	\$144,000	\$450,000
Pro revenue	\$12,000	\$48,000	\$144,000
Total revenue	\$78,000	\$282,000	\$774,000
Voice and card COGS	(\$22,104)	(\$83,232)	(\$240,444)
Free tier (marketing)	(\$4,320)	(\$21,600)	(\$64,800)
Gross profit	\$51,576	\$177,168	\$468,756
Gross margin	66%	63%	61%

Margin declines by design: campus is lower-margin than cohort and grows as a share of the mix. Figures are gross profit and exclude salaries and sales costs.

The AssemblyAI credits fund year one. \$5,000 is 66,000 session minutes - about 22,000 free questions, covering our entire year-one free tier at \$4,320 of modelled consumption.

Risks and mitigations

Risk	Mitigation
High adoption compresses campus margin	Pooled allowance plus \$0.15/min overage; idle-timeout tiering cuts unit cost 58%
Campus sales cycles run 3-9 months	Bootcamps carry year-one revenue; campus sold course by course, avoiding procurement
GitHub Copilot is free for students	Different job, different buyer: Copilot writes code, we explain code already written
Voice is a genuine cost of goods	Allowances capped in code; margin is 60-83% and we say so

What must be validated next

- 1. Questions per student per week.** Every figure in this plan hangs off this number. One teaching term of usage logs settles it, and the product already records each turn.
- 2. One real price quote.** A single bootcamp or lecturer stating what they would pay per learner is worth more than any projection on these pages.
- 3. Adoption rate per seat.** Determines whether a campus deal earns 77% or 20%, and therefore whether the overage term is commercial protection or the main revenue line.